Ziyang Zhang

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Ph.D. student at Tsinghua EE. Research: time-series LLMs & world models, speech generation (TTS, speech LLMs), LLM alignment (DPO, weak-to-strong).

Education

Tsinghua University — Beijing, China
Ph.D. in Electronic Engineering (Advisor: Assoc. Prof. Chao Zhang) — *Aug. 2024 – Present*

Tsinghua University — Beijing, China
B.Eng. in Electronic Engineering; Outstanding Graduate (2024); Academic Excellence Scholarship ×3 — *Sept. 2020 – July 2024*

Research Internships

Pujiang National Laboratory — Nov. 2024 – Present
Research Intern, Time-Series Foundation Models (Intern-S series time-series team)

- **Intern-S1 / Intern-S2-Preview**: owned the overall architecture and training (model integration) of the time-series modality, performing cross-modal alignment between the time-series encoder and the LLM, followed by SFT; independently led the architecture exploration for the time-series forecasting capability newly added in Intern-S2.
- **MMTA & TimeOmni-v** (first author, **NeurIPS 2026**): built a time series + text + vision benchmark (3,524 test samples, 9 domains); showed that serializing series as text is the shared bottleneck of 8 MLLMs; designed TimeOmni-v with a TS encoder and temporal positional alignment.
- **SciTS & TimeOmni** (second author, **ICLR 2026**): solely responsible for the architecture design and training of TimeOmni, a unified time-series LLM for both understanding and generation; a length-based router dispatches each signal to one of several time-series encoders, so TimeOmni handles diverse series spanning 100 to 10M points, with a forecasting head for generation; evaluated on SciTS (12 domains, 43 tasks, 54k instances).
- **TimeOmni-o1** (first author): coupled a pretrained time-series foundation model and an LLM in both directions; the model reasons in text first, then passes its reasoning back to the TSFM so forecasts respond to event text; built MMTR with annotated reasoning traces (500k+ samples, 12 domains); beats prior models on understanding and forecasting, also on three open benchmarks.
- **Time-JEPA** (co-first author): supervised video world models with temporally aligned physical time series, with no hand-coded physical laws; the model fits physical laws better (more accurate kinematic reconstruction), keeps its gains under unseen conditions and unseen state variables, generalizes to new scenarios, and is more sensitive to physical anomalies (driving, robot manipulation, human motion).

Apple, Siri China R&D — Jan. 2024 – Oct. 2024
Research Intern, Speech Generation

- **BELLE** (first author): one text can be spoken in many valid ways (a one-to-many problem). To model this uncertainty, proposed BELLE, which brings Bayesian evidential learning into continuous-valued autoregressive TTS: it models the acoustic target as a Normal-Inverse-Gamma distribution to capture data-dependent aleatoric uncertainty without extra parameters or inference latency, and uses a one-to-many training strategy with synthetic samples as a statistical support set. Trained on ∼5k hours, it achieves a 25.8% relative WER reduction and supports high-quality streaming generation.

Other Research

Speech LLMs

- **TVPara** (first author): built a 700h+ paralinguistic captioning dataset from TV series; fine-tuning open-source LALMs on it improves paralinguistic understanding in complex acoustic environments by 31.26% (relative).

LLM Alignment

- **Bayesian WeakS-to-Strong** (co-first author, **ICLR 2025**): extended weak-to-strong generalization to an ensemble of weak supervisors modeling diverse human opinions, with Bayesian confidence estimates guiding the strong model; moved from classification to generation via DPO-based preference learning.

Publications

** equal contribution*

[1] **Ziyang Zhang***, S. Li*, Y. Wang, Z. Cui, B. Zhou, W. Wu, C. Zhang. MMTA: Benchmarking Multimodal Temporal Analysis with Time Series, Text, and Vision. **NeurIPS 2026**. arXiv

[2] W. Wu, **Ziyang Zhang**, L. Liu, X. Xu, et al., C. Zhang. SciTS: Scientific Time Series Understanding and Generation with LLMs. **ICLR 2026**. paper

[3] Z. Cui*, **Ziyang Zhang***, G. Sun, W. Wu, C. Zhang. Bayesian WeakS-to-Strong from Text Classification to Generation. **ICLR 2025**. paper

[4] **Ziyang Zhang**, A. Thwaites, A. Woolgar, B. Moore, C. Zhang. SWIM: Short-Window CNN Integrated with Mamba for EEG-Based Auditory Spatial Attention Decoding. **IEEE SLT 2024**.

[5] **Ziyang Zhang**, S. Li, W. Wu, C. Zhang. TimeOmni-o1: Bidirectionally Coupling Time Series Foundation Models and LLMs for Multimodal Reasoning and Forecasting. Under review. OpenReview

[6] **Ziyang Zhang** et al. (co-first author). Time-JEPA: Grounding Video World Models with Physical Time Series. Under review. OpenReview

[7] **Ziyang Zhang**, W. Yu, W. Wu, X. Xu, C. Zhang. TVPara: Paralinguistic Captioning of Expressive TV-Series Speech in Complex Acoustic Environments. arXiv 2026. arXiv

[8] **Ziyang Zhang**, Y. Gao, X. Xu, B. Li, W. Wu, C. Zhang. Bayesian Speech Synthesizers Can Learn from Multiple Teachers. arXiv 2025. arXiv

[9] Intern-S2 Team (incl. **Ziyang Zhang**). Intern-S2-Preview: Scientific Agentic Foundation Model. Technical report, 2026. arXiv

[10] Intern-S1 Team (incl. **Ziyang Zhang**). Intern-S1: A Scientific Multimodal Foundation Model. Technical report, 2025. arXiv